

THE UNITED REPUBLIC OF TANZANIA
NATIONAL EXAMINATIONS COUNCIL
ADVANCED CERTIFICATE OF SECONDARY EDUCATION

EXAMINATIONS

131/3A

PHYSICS 3A
(PRACTICAL A)

(For Both School and Private Candidates)

Duration: 3:20 Hours

Year: 2025

Instructions

1. This paper consists of **three (3)** questions.
2. Answer **all** questions.
3. Question **one (1)** carries **twenty (20)** marks and the other **two (2)** carry **fifteen (15)** marks each.
4. Mathematics tables and non-programmable calculators may be used.
5. All writing must be in **black** or **blue** ink except for drawings which must be in pencil
6. Communication devices and any unauthorised materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer booklet(s).

The following information may be useful:

Pie, $\pi = 3.14$

Specific heat capacity of water = $4.2 \times 10^3 \text{ Jkg}^{-1}\text{K}^{-1}$



1. You are provided with the plastic ruler (30 cm), metre rule, retort stand with its accessories, mass, **M**, thread, masking tape, micro meter screw gauge and weighing balance. Proceed as follows:

- (a) By using cork pads on the retort stand, clamp the given 30 cm ruler so that its flat side is horizontal and the protruding length, $l = 27$ cm. Using a masking tape, fix the thread with length $L = 110$ cm at the end of the ruler and the mass, **M** hanging at the bottom as shown in Diagram 9.

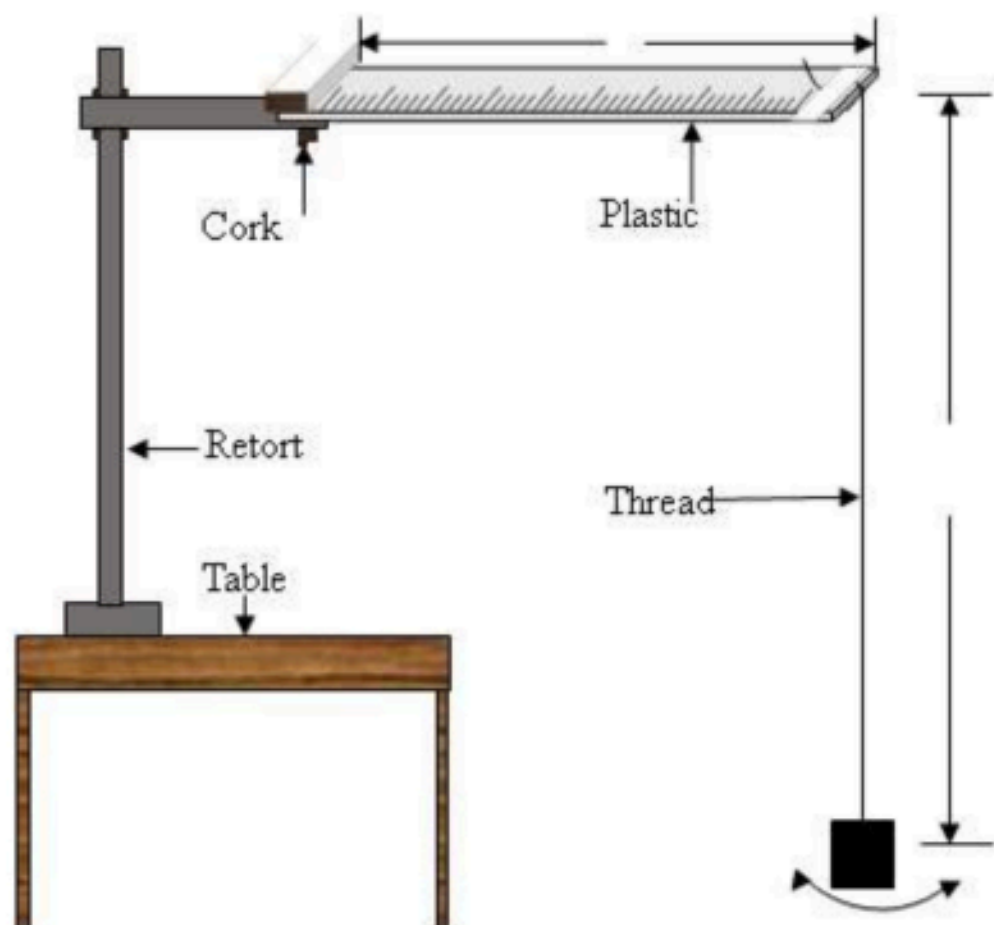


Diagram 6

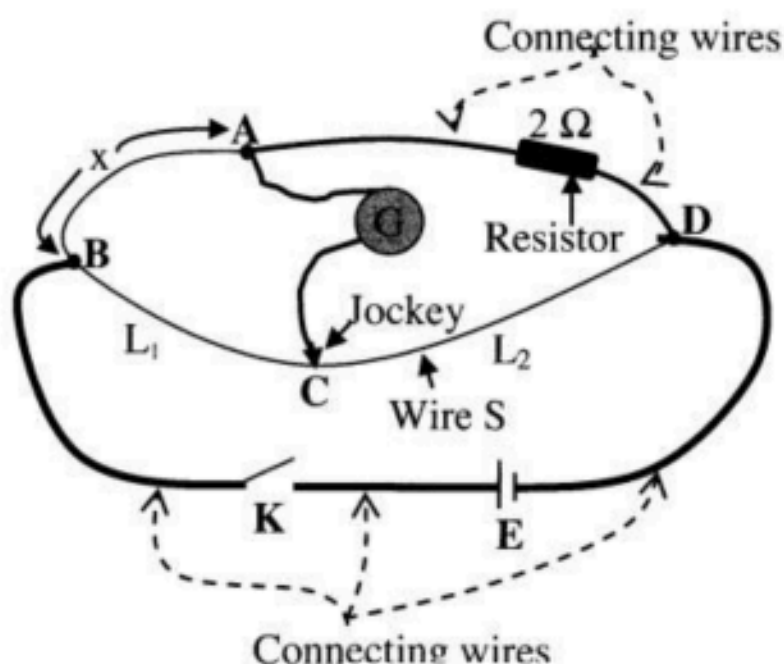
- (b) Displace the mass, **M** a small distance from its equilibrium and release it so that it executes oscillations in a plane parallel to the ruler. Measure and record the time, t_{10} for ten oscillations and hence compute its periodic time T .
- (c) Repeat the procedures in 1 (b) when the lengths of the thread is $L = 90$ cm, 70 cm, 50 cm and 30 cm
 - (i) Tabulate your results including the values of L (m), t_{10} (s), T (s) and T^2 (s²)
 - (ii) Plot a graph of T^2 (s²) against L (cm)
 - (iii) If T and L are related by $T^2 = 4.03L + T_c^2$, what is the value of T_c^2 ?
 - (iv) State the physical meaning of T_c
 - (v) Measure and record the mass of the solid, **M** and the breadth, b and thickness, t of the ruler.
 - (vi) Compare the Young's modulus E of the plastic ruler in SI, given that

$$E = \frac{16\pi^2 M}{bT_c} \left(\frac{l}{t}\right)^3$$

2. You are required to investigate the cooling behavior of a copper calorimeter under different conditions. Proceed as follows:
 - (a) Half-filled the calorimeter with hot water of about 90°. Cover the calorimeter with a lid and insert the thermometer through the opening so as to read the temperature of water.
 - (b) Starting with temperature of 80°, read and record the temperature of water at the interval of one minute for 10 minutes while stirring and fanning using cardboard.
 - (c) Wet the given cloth normal tap water.
 - (d) Repeat the procedures in 2 (a) and (b), but in this case wrap the calorimeter with a wet cloth just before starting recording the temperature.

Questions:

- (i) Tabulate your results obtained in 2 (b) and (d).
 - (ii) Plot the cooling curves for the results in 2 (b) and (d) on the same plane of axes.
 - (iii) Determine the temperature of water at 7th minute for both cooling curves.
 - (iv) In which of the two settings did water cools faster? Justify the answer by giving two points.
3. You have been provided with a $2\ \Omega$ standard resistor, resistance wire **S**, galvanometer **G**, dry cell **E** (1.5 V, size D), switch **K**, jockey and connecting wires. Determine the resistivity of wire, **S** through the following procedures.
- (a) Connect 80 cm of wire **S** together with $2\ \Omega$ resistor in series to make a loop.
 - (b) Connect wires from the terminals of the dry cell **E**, at junction **B** and junction **D**, then connect resistance wire **S** from the junction **A** to junction **B** and complete the circuit as shown in a Diagram 9.



- (c) Close the switch, K and determine the balancing point C. Read and record the lengths, L_1 and L_2
- (d) Repeat the procedures in 3(b) and (c) for the value of x equal to 20cm, 30cm, 40cm, and 50cm.

Questions:

- (i) Tabulate your results including the value of x, L_1 , L_2 and $\frac{L_1}{L_2}$
- (ii) Plot a graph of x (cm) against $\frac{L_1}{L_2}$
- (iii) Determine the slope of the graph.
- (iv) Measure and record the diameter of the wire, S by using micro meter screw gauge.
- (v) Determine the resistivity of a wire, S using the answer obtained in 3 (iii) and (iv).